

Neural Network and Deep Learning

Total Time: 3 Hours

Theme: DL Problem Identification & Workflow

Dataset: [Fashion MNIST \(image classification\)](#)

Section A: Concept Application

1. Fashion MNIST contains raw 28x28 pixel images. Why is Deep Learning more suitable than traditional ML for classifying raw image data?
2. The DL workflow follows a fixed sequence of stages. List the stages of the Deep Learning workflow in order.
3. A biological neuron fires only when incoming signals exceed a threshold. What is the artificial equivalent of this threshold mechanism in a neural network?
4. You must choose between an ANN and a CNN for Fashion MNIST. What property of CNNs makes them better suited for image data than a plain ANN?
5. Deep Learning was theorized decades ago but became practical only recently. Name two factors that enabled the rise of modern Deep Learning.

Section B: Practical Task

1. Load the Fashion MNIST dataset and normalize pixel values to a range of $[0, 1]$ using NumPy or TensorFlow.
2. Implement a baseline Artificial Neural Network (ANN) with at least two dense layers to establish a performance benchmark.
3. Design a Convolutional Neural Network (CNN) architecture featuring Conv2D, MaxPooling2D, and Dropout layers for spatial feature extraction.
4. Execute a comparative training run for 10 epochs and plot the training vs. validation accuracy to identify potential overfitting.

Section C: Mini Project

1. **Title:** Fashion-Forward: Automated Inventory Categorization System
2. **Problem Statement:** Develop an automated visual classification pipeline to reduce manual tagging errors in a high-volume e-commerce warehouse.
3. **Dataset Recommendation:** [Fashion MNIST \(image classification\)](#)
4. **Required Deliverables:** Preprocessing script, CNN Model Architecture Code, Accuracy/Loss Visualization Plots, Model Evaluation Report (Confusion Matrix).